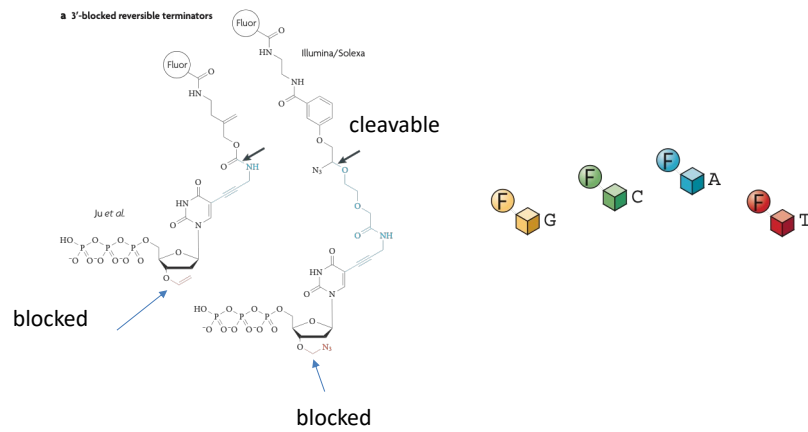


2. Illumina Sequencing Sequencing By Synthesis (SBS)



- Terminated through reversible 3' terminators
- 4-color system
- Fluorescent nucleotides are imaged one at a time
- Single-strand DNA is immobilized

Mardis, 2008 &
Metzker, 2010

20

2. Illumina Sequencing Sequencing By Synthesis (SBS)

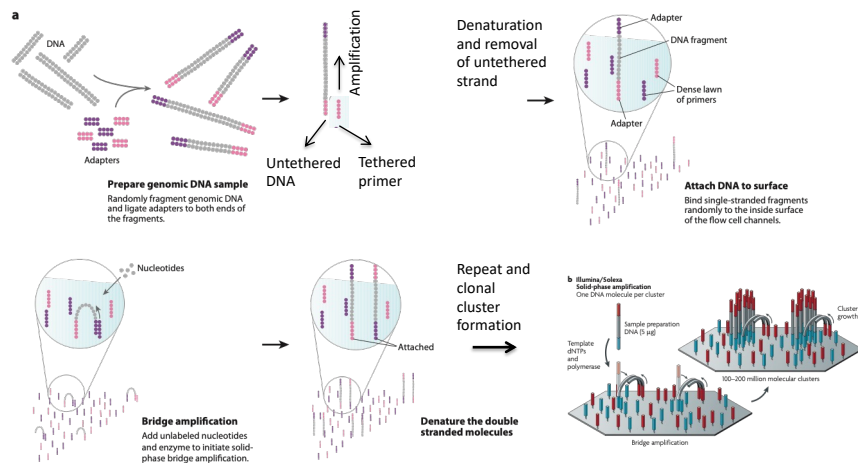


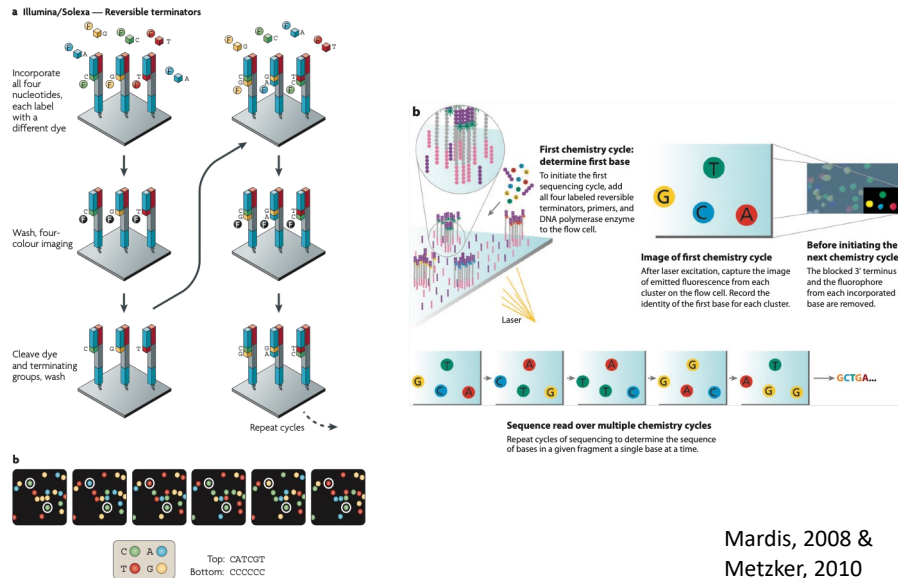
Figure 2

The Illumina sequencing-by-synthesis approach. Cluster strands created by bridge amplification are primed and all four fluorescently labeled, 3'-OH blocked nucleotides are added to the flow cell with DNA polymerase. The cluster strands are extended by one nucleotide. Following the incorporation step, the unused nucleotides and DNA polymerase molecules are washed away, a scan buffer is added to the flow cell, and the optics system scans each lane of the flow cell by imaging units called tiles. Once imaging is completed, chemicals that effect cleavage of the fluorescent labels and the 3'-OH blocking groups are added to the flow cell, which prepares the cluster strands for another round of fluorescent nucleotide incorporation.

Mardis, 2008 &
Metzker, 2010

21

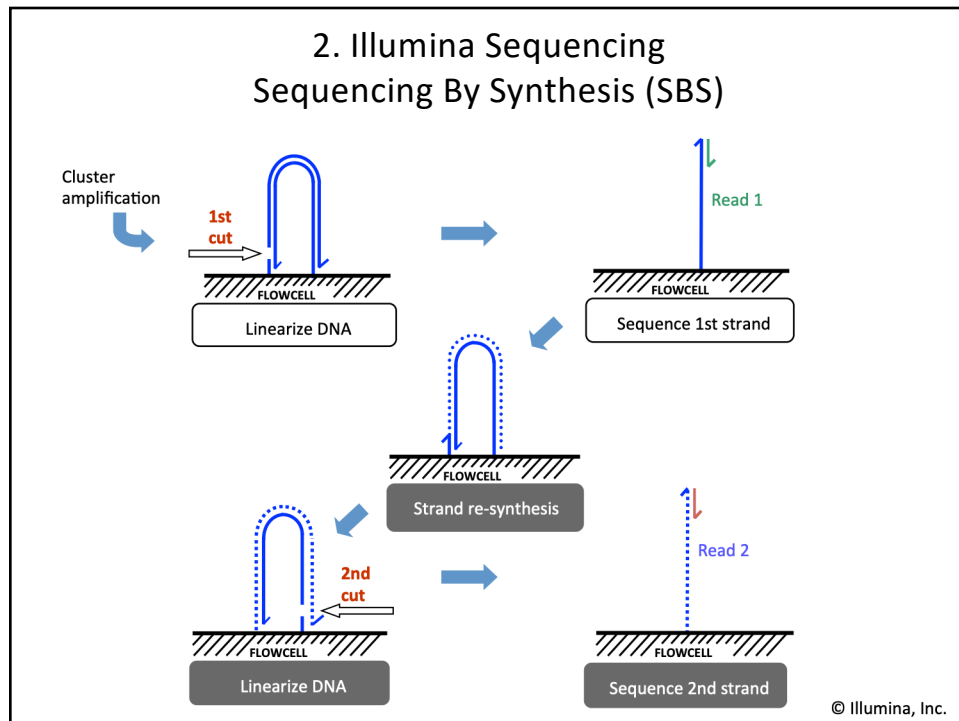
2. Illumina Sequencing Sequencing By Synthesis (SBS)



22

- Watch this Illumina video for more details:
<https://www.youtube.com/watch?v=fCd6B5HRaZ8>

23



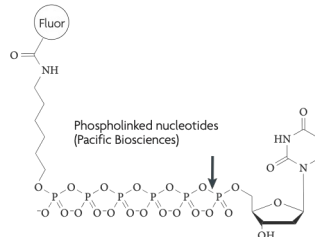
24

2. Illumina Sequencing Sequencing By Synthesis (SBS)

- Less accurate – higher error rate
- Short reads (100-200 bps)
- Less expensive
- Paired-end
- Fast
- Requires very little amount of DNA (2 ng)
- Easy sample preparation – no cloning needed
- VERY high-throughput

25

3. Single Molecule Real Time SMRT Sequencing (Pacific Biosciences)

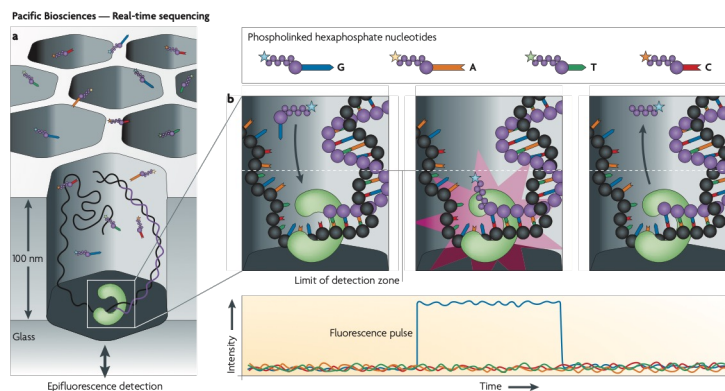


- The fluorophores are on the phosphate group, and are cleaved during DNA synthesis
- 4-color system
- Fluorescent nucleotides are imaged one at a time as they are bound to the polymerase during synthesis
- Polymerase-template complex is immobilized

Mardis, 2008 &
Metzker, 2010

26

3. Single Molecule Real Time SMRT Sequencing (Pacific Biosciences)



Detection in 20-30
nm (10^{-21} L)

Mardis, 2008 &
Metzker, 2010

Figure 4 | **Real-time sequencing.** Pacific Biosciences' four-colour real-time sequencing method is shown. **a** | The zero-mode waveguide (ZMW) design reduces the observation volume, therefore reducing the number of stray fluorescently labelled molecules that enter the detection layer for a given period. These ZMW detectors address the dilemma that DNA polymerases perform optimally when fluorescently labelled nucleotides are present in the micromolar concentration range², whereas most single-molecule detection methods perform optimally when fluorescent species are in the pico- to nanomolar concentration range². **b** | The residence time of phospholinked nucleotides in the active site is governed by the rate of catalysis and is usually on the millisecond scale. This corresponds to a recorded fluorescence pulse, because only the bound, dye-labelled nucleotide occupies the ZMW detection zone on this timescale. The released, dye-labelled pentaphosphate by-product quickly diffuses away, dropping the fluorescence signal to background levels. Translocation of the template marks the interphase period before binding and incorporation of the next incoming phospholinked nucleotide.

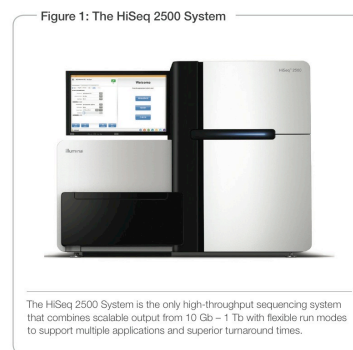
27

3. Single Molecule Real Time SMRT Sequencing (Pacific Biosciences)

- Less accurate – higher error rate
- Very long reads (up to 30, 000 bps)
- Expensive
- Fast
- Easy sample preparation – no cloning needed,
- Requires high quality and large amount of DNA (20 ug)
- Higher throughput than Sanger but lower than Illumina

28

An Illumina machine gives you raw reads



What to do with millions of reads?

29

Formats and Quality Scores

1. Fasta format: **>short_name** long name SEQUENCE

```
[donia@molbio5 RNA-Seq]$ more subX319146421 Stool RNA.fa
>SRR769401.1.1 1 length=101
TTTTTCGATGTGCCTTTTGGCAAGGTGAATGTGGGTACAGAAATATACATCTATGCCTAAATCTTCTCGAGTATTTATGTTCTGACATCTCGCGTCC
>SRR769401.1.2 1 length=101
GACAATAAAGTTTCGATAGCAACCTTATTGAACGAAATACACGGCTCTGTATCTTGGCAACATTACCTGATGCAAGGCAGAAATCAGTGCAGCAAGGCTTT
>SRR769401.2.1 2 length=101
CATCTAGTTGGGCTTTTTCATCCGAAAAGTTCAGAGTGAGATCTTATCTGGCAAGGATTAATTCATTATTTTCCTTGCCACGTCCCAAATTAACAAT
>SRR769401.2.2 2 length=101
CCAAATGAATTAAGGTTCAAGCGGAGCGAATTCTAAAGAAGTGGGTTAAATACTACAGATGCTATTGCTTATTATTAACGCAAGTAGTAAATCGAGG
>SRR769401.3.1 3 length=101
TAGCTTTTCAGTGCCTGAGAGTATTTCCCATTTTCAGGTAATTCGAAAGCTTGTAGCAAATAAGTGGCGTTGAGCATCTGTTATTCTGAAAAAATATC
>SRR769401.3.2 3 length=101
GTTGAACAGCATAAAGAGGTCGCTCTATTCTGGGTTCTCTTTAGCTTCAAACAAAAAGAAATCCGAACAAGCAATAGAACGTAAGAAAGATCCCAA
>SRR769401.4.1 4 length=101
GCCGGTATGTACCGGCTCCCAATCCATTAAATTGACTTAACAGTACAGTAAAGATAGTGCTTGCCAAAAGAGATCCTGACTATTGTGTCTAAGGGAAAA
>SRR769401.4.2 4 length=101
TATTGGCGCTCATTGGCGTTTTCGACCTGATTACTCATTTCAATAGTCGATTCTATGTCTTATGTAATAATGTGGCTGAAATCGCATGTGAGAGCGTTT
>SRR769401.5.1 5 length=101
GTAAATAGAGATAAGACATCTCAATTTTAGAGTTTCTTTAGGTATGATATAAAAAATACAAAGCCTAACAGTTAGTTTAAAGGTTAAAAATTGAA
>SRR769401.5.2 5 length=101
CCAAACTTCAGCTTGATTTTCATCTGAATCAACCATGACCATATAGATAATCTCGGCTAATCTCACTAGCAGAGATATCTCCATTTCTGCTTTG
>SRR769401.6.1 6 length=101
GCAGGAGCTGGATCTGTAAGTCATCGGCAGGTACGTATACAGCTGAACCGATGGGATAGCACCAGCCTTTGTAGATGTGATACGTTCTGGAAGTGAAC
```

30

Formats and Quality Scores

1. Fasta format: **>short_name** long name SEQUENCE

-- Used for raw reads, assembled contigs and complete genomes

```
>gi|113200490|gb|J01415.2|HUMMTCG Homo sapiens mitochondrion, complete genome
GATCACAGGTCTATCACCTTATTAACTCAGCGGAGCTCTCCATGCTTTGGTATTTTCGCTGGGGG
GTATGCACGCGATGATGCTTGCAGACGCTGGAGCCGAGCACCTATGTCGAGTATCTGTCTTTGATT
CTGCCTCATCTTATTATTATCGACCTACGTTCAATATTACAGGCGAACATACTACTAAAGTGTGTTA
ATTAATTAAGTCTGTAGACATAATAACAATGAATGCTGTCAGACGCCACTTCCACACAGACATC
ATAACAAAAATTTCCACAAACCCCTCCCGCTTCTGGCCACAGCACTTAAACACATCTCTGCCA
AACCCAAAAACAAAGAACCTTAACACGAGCTAACAGATTTCAAATTTTATCTTTGGCGGTATGCAC
TTTTAACAGTCACCCCACTAACACATTATTTCCCTCCCACTCCCACTACTAATCTCATCAATA
CAACCCCGCCATCTTACCCAGCACACACACCGCTGCTAACCCCATACCCGAACCAACCAACCC
AAAGACACCCCCACAGTTTATGTAGCTTACCTCTCAAAGCAATCACTGAAATGTTTAGACGGGCTC
ACATACCCCAATAAAGATAGTTTGGTCTAGCCTTCTATTAGCTCTTAGTAAGATTACACATGCAA
GCATCCCGTTCAGTGAGTTTACCCTTAAATCACACGATCAAAAGGAACAAGCATCAAGCAGCGAGC
AATGCAGCTCAAAACGCTTAGCCTAGCCACACCCACGCGGAAACAGCAGTGATTAACCTTAGCAATAA
ACGAAAGTTTAACTAAGCTATACTAACCCAGGGTTGGTCAATTTCTGTCAGCCACCGCGGTACACGA
TTAACCAAGTCAATAGAACCGGCGTAAAGAGTGTGTTAGATCACCCCTCCCAATAAAGCTAAACT
CACCTGAGTTGTAATAAAGTCCAGTTGACACAAAATAGACTACGAAAGTGGCTTAAACATATCTGAACAC
ACAATAGCTAAGACCCAACTGGGATTAGATACCCCACTATGCTTAGCCCTAAACCTCAACAGTTAAATC
AACAAAACCTGCTGCCAGAACACTACGAGCCACAGCTTAAACTCAAAGGACCTGGCGGTGCTTCATATC
CCTCTAGAGGAGCCTGTTCTGTAATCGATAAACCCGATCAACCTCACCACTCTTGCTCAGCTATATA
CCGCATCTTCAGCAACCTGATGAAGGTACAAAGTAAGCGCAAGTACCCAGTAAAGACGTAGGTC
```

31

NTNPQS_EpoB
MTNLQSLNELEHQGVKLAAADGERLQIQAPKNALPNLLARISEHKSTILTMLRQRLPAESIVPAPAEHRV
PPLTLDIGQSYWLGRGTAFVTPSGHIAHEYDCTDLVRLARSFRKRVARHDEQMLRAHTLPDMMQVIEKP
VDADIEIHLDLRDSTREARLVSDHIMSHYDTERPPIYHVAVRLVDEQTRLVSLDILNDVGLSI
SIIFKDWLSFYEDPETSPLVLELSYRDYVLALESRKKEASHQRSMDYWKRRVAELPPPPMLPMKADPSTL
RIERFRHTEQWLPSDWSRQKQVGERGLTPTGVILAESEVIGRWASAPRFLTNILNRLPVPVRVND
ITGDFSVMLLDITDTRKFSQEARKRIQQLWEAMHDCVDSGEVQREAEJQIRGALFPVLVPT
NQQVQVMSLTLQRLGRTPVYTSQTQLLLDQLHQEYHGDGLVLAIDVQVFPDPLDLMKYLVAEFLRRLT
EPEEVCGRCTSLPAQLEARRASTENTSLSETHLGFARVEAQVPMQLAVNSDKARTYEELSRSSRL
LGRLEEQGARQTLVSVMEKGEVQDLVAVLAVESGAQVVPIDALPAERHIVLHDGEVKVLTPQWPL
DGLSWSPQIRTLVSDAVGEGDQQLPMMPIQTSPLDAYIYVTSGSTGLPKGYMHDRHAGVNTILNDIE
RFIEGPDRVLVLSLSDFLSYVDFGILAAVGLQVDPADSKLRDPASHVAHEIERKTVYVNSVPLMRM
VLEHSGRPDSLARSRLLSGSDVPIVGLPGLGELQVAPVLSGGATSEIWSIGYVPRNVDSMWS
IPYGRPLRNQTFHVLDEALEPRPVVWPGQLYIGGVGLALGYWRDEEKTNSFLVHPETGERLYKTGDLGR
YLPDGNIEFMGDREAHQIKRLGYVSELGIEVLTSLKSPNVRDIAVPVGNDAKALLNYVPEGTRRRA
EQDASLTERVADNAAHAKADGLSDGERVQFKLARHRLRDLQDGVPTDLVGRPAGLDVYARRRSRV
TLEAIPFIYVGRFLCSLSSVPEGAALPKFPRYSAGSTSYVPTAYAKSGRIEGVDEAGMYLFEFHR
LKVSDHGIERGHAVPQNFVDAAEAALFVGRDIAESYLSRFFCELLEAGQGLLMEQAPSCNI

2. Fastq format: @**read_identifier** paired end info
SEQUENCE
+(**read_identifier** paired end info)
QUALITY SCORES

```

@JLK5VL1:806:HJVTKBCXX:2:1101:20635:7461 1:N:0:
AATCCATTGGCTAATTTAGTTCGTTGGAAAGATAAACCACTAGCCTTATCTATTGTCCAAGCTCGA
+
DDDDDIHHIIHHIIHHIIHHIIHHIIHHIIHHIIHHIIHHIIHHIIHHIIF
@JLK5VL1:806:HJVTKBCXX:2:1101:1314:7716 1:N:0:
ACTAACGCTGGTTACAAAATTTGGTCAGGAACCTGAACTTACAACATCGTTGCTGCACATGGTTATT
+
DADDHIIHHHHHHHHHIIICEHEHHIIDHHIGCGHIEEHHIIHHIHHHHHHHHI?HHEG?GHH
@JLK5VL1:806:HJVTKBCXX:2:1101:2048:7504 1:N:0:
AATCAGCATTAAGTTCGTACTCTTTTATTATAACTGTTTGAGTTTCTGAATACTGTGGGTTAG
+
DDDDDIHHIIHHIIHHIIHHIIHHIIHHIIHHIIHHIIHHIIHHIIHHIIF
@JLK5VL1:806:HJVTKBCXX:2:1101:3207:7656 1:N:0:
TAATCACGGGTGCCCTCGCTGTGAGGGCTCAATGGCACATTGTTATGCACCACACCTCGAGCTTCAGG
+
@@DADGHIIHHIHHIHHIHHFHDEH=?GEHCCHIIHHIIGHHHIIHHIEDHICDHENHHHIFFG
@JLK5VL1:806:HJVTKBCXX:2:1101:3918:7719 1:N:0:
GAATCAGCATCTGTAGTTATTATTTCTCAAATTTTGCTTCCAAAGATTAAATATCTCTGTCTCAA
+
DDDDDIHHIIHHIIHHIIHHIIHHIIHHIIHHIIHHIIHHIHEWHIIHGIIHHIIGHHHIIHH

```


How are quality scores calculated?

- Quality values represent the probability that the nucleotide was **incorrectly** called at that position. If the sequencer/software called a nucleotide as a **T**, the quality value at that position represents the probability that this nucleotide is **NOT a T**.
- This probability is rescaled using the Phread scale (a negative log scale).

36

How are quality scores calculated?

$$Q_{\text{value}} = -10 \log_{10} p$$

$$\begin{aligned} \text{So, if your } p=0.1, \text{ then } Q_{\text{value}} &= (-10 \log_{10}(0.1)) \\ &= (-10(-1)) = 10 \end{aligned}$$

$$\begin{aligned} \text{If your } p=0.01, \text{ then } Q_{\text{value}} &= (-10 \log_{10}(0.01)) \\ &= (-10(-2)) = 20 \end{aligned}$$

$$\begin{aligned} \text{If } p=0.001, \text{ then } Q_{\text{value}} &= (-10 \log_{10}(0.001)) \\ &= (-10(-3)) = 30 \end{aligned}$$

37

How are quality scores calculated?

- Quality values are then rounded up to the closed integer, and represented as a single bit in the American Standard Code for Information Interchange format (ASCII).
- For Phred33, the first 33 characters are control characters and we start after them. (there is also Phred64)

Example:

For $Q=30$, use ASCII character of $33+30=63$, which is “?”

For $Q=20$, use ASCII character of $33+20=53$, which is “5”

ASCII printable code chart [edit]

Binary	Oct	Dec	Hex	Glyph	Binary	Oct	Dec	Hex	Glyph	Binary	Oct	Dec	Hex	Glyph
010 0000	040	32	20	(space)	100 0000	100	64	40	@	110 0000	140	96	60	~
010 0001	041	33	21	!	100 0001	101	65	41	A	110 0001	141	97	61	a
010 0010	042	34	22	"	100 0010	102	66	42	B	110 0010	142	98	62	b
010 0011	043	35	23	#	100 0011	103	67	43	C	110 0011	143	99	63	c
010 0100	044	36	24	\$	100 0100	104	68	44	D	110 0100	144	100	64	d
010 0101	045	37	25	%	100 0101	105	69	45	E	110 0101	145	101	65	e
010 0110	046	38	26	&	100 0110	106	70	46	F	110 0110	146	102	66	f
010 0111	047	39	27	'	100 0111	107	71	47	G	110 0111	147	103	67	g
010 1000	050	40	28	(	100 1000	110	72	48	H	110 1000	150	104	68	h
010 1001	051	41	29	)	100 1001	111	73	49	I	110 1001	151	105	69	i
010 1010	052	42	2A	*	100 1010	112	74	4A	J	110 1010	152	106	6A	j
010 1011	053	43	2B	+	100 1011	113	75	4B	K	110 1011	153	107	6B	k
010 1100	054	44	2C	,	100 1100	114	76	4C	L	110 1100	154	108	6C	l
010 1101	055	45	2D	-	100 1101	115	77	4D	M	110 1101	155	109	6D	m
010 1110	056	46	2E	.	100 1110	116	78	4E	N	110 1110	156	110	6E	n
010 1111	057	47	2F	/	100 1111	117	79	4F	O	110 1111	157	111	6F	o
011 0000	060	48	30	0	101 0000	120	80	50	P	111 0000	160	112	70	p
011 0001	061	49	31	1	101 0001	121	81	51	Q	111 0001	161	113	71	q
011 0010	062	50	32	2	101 0010	122	82	52	R	111 0010	162	114	72	r
011 0011	063	51	33	3	101 0011	123	83	53	S	111 0011	163	115	73	s
011 0100	064	52	34	4	101 0100	124	84	54	T	111 0100	164	116	74	t
011 0101	065	53	35	5	101 0101	125	85	55	U	111 0101	165	117	75	u
011 0110	066	54	36	6	101 0110	126	86	56	V	111 0110	166	118	76	v
011 0111	067	55	37	7	101 0111	127	87	57	W	111 0111	167	119	77	w
011 1000	070	56	38	8	101 1000	130	88	58	X	111 1000	170	120	78	x
011 1001	071	57	39	9	101 1001	131	89	59	Y	111 1001	171	121	79	y
011 1010	072	58	3A	:	101 1010	132	90	5A	Z	111 1010	172	122	7A	z
011 1011	073	59	3B	;	101 1011	133	91	5B	[	111 1011	173	123	7B	{
011 1100	074	60	3C	<	101 1100	134	92	5C	\	111 1100	174	124	7C	
011 1101	075	61	3D	=	101 1101	135	93	5D	]	111 1101	175	125	7D	}
011 1110	076	62	3E	>	101 1110	136	94	5E	^	111 1110	176	126	7E	~
011 1111	077	63	3F	?	101 1111	137	95	5F	_					

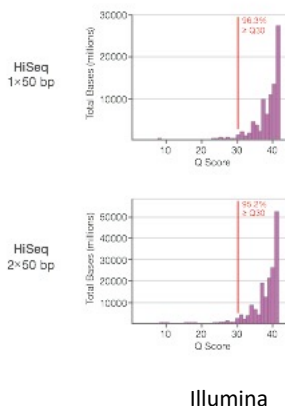
38

Quality Filtering

- Important for the accuracy of the underlying application (assembly, mapping, etc.)

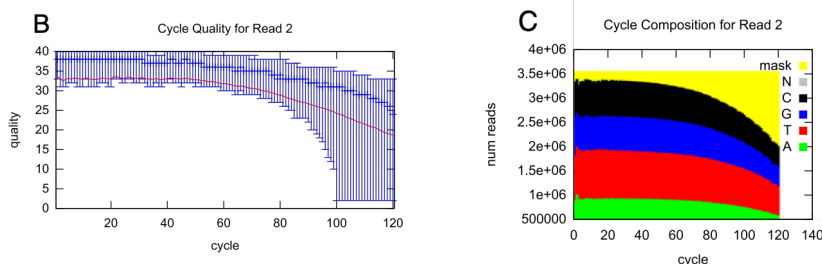
Table 1: Quality Scores and Base Calling Accuracy

Phred Quality Score	Probability of Incorrect Base Call	Base Call Accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1,000	99.9%
40	1 in 10,000	99.99%
50	1 in 100,000	99.999%



39

Quality Filtering vs trimming



Where do errors and low quality scores come from?

1. PCR errors (homopolymeric regions)
2. Dark nucleotides
3. Low sequence diversity
4. De-phasing in the clusters

Yang, et al, HTQC: a fast quality control toolkit for Illumina sequencing data, BMC Bioinformatics, 2013

40

PrinSEQ (<http://prinseq.sourceforge.net/manual.html>)

BIOINFORMATICS APPLICATIONS NOTE Vol. 27 no. 6 2011, pages 863–864
doi:10.1093/bioinformatics/btr026

Sequence analysis

Advance Access publication January 28, 2011

Quality control and preprocessing of metagenomic datasets

Robert Schmieder^{1,2,*} and Robert Edwards^{1,3,*}

¹Department of Computer Science, ²Computational Science Research Center, San Diego State University, San Diego, CA 92182 and ³Mathematics and Computer Science Division, Argonne National Laboratory, Argonne, IL 60439, USA

Associate Editor: Alex Bateman

-- Quality filtering, trimming, and paired-end handling of sequencing data

Example:

I want to make sure that **every remaining read is at least 70 bp long, with an average probability of 99.9% chance of being correct, after trimming the ends at the same quality, and removing ones that have more than two Ns.**

```
nohup prinseq-lite.pl -fastq read1.fastq -fastq2 read2.fastq -
out_good Run1-good -out_bad Run1-bad -trim_qual_left 30 -
trim_qual_right 30 -min_len 70 -min_qual_mean 30 -ns_max_p 2 -
trim_qual_type mean -trim_qual_window 10 -verbose -log
prinseq.1.log &
```

42